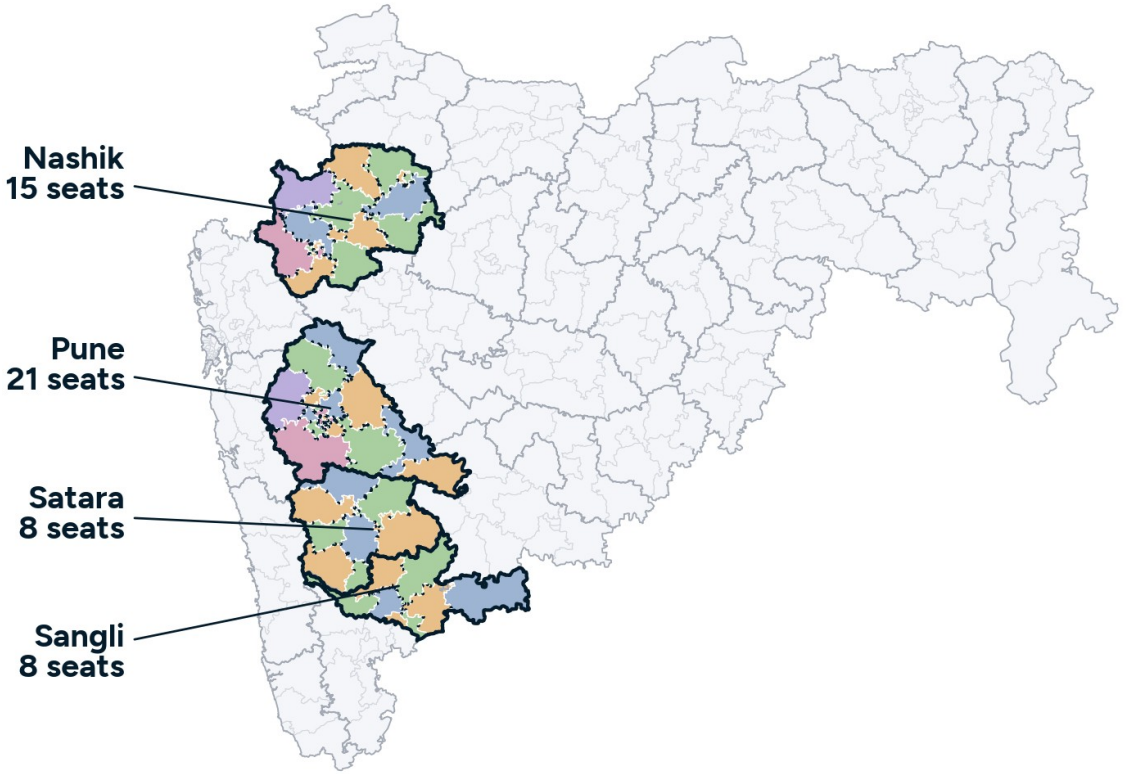
IE301 OPERATIONS ANALYSIS · GROUP 4 · FINAL PRESENTATION

Election Logistics

Pari Soni (24b4505) · Shivam Soni (24b4525)
Shreyaan Sharma (24b4529)
Asmaniwala Devanshu (26x0053) · S Ganesh (26x0052)

Mentor: Prof. Ketan Savla

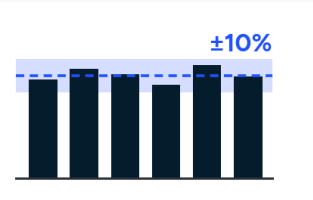


Problem Statement

How should a district be split into Assembly seats so every seat holds a fair share of people, and where should polling stations go so every voter can reach one on foot?

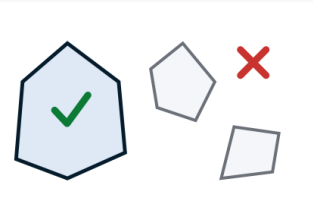
1 Constituency design

Where do the boundaries go?



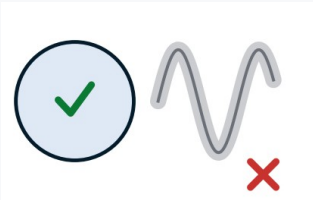
Population balance

every seat within ±10% of the district average



Contiguity

each seat is one connected piece

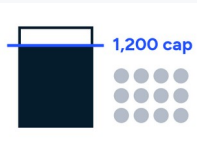


Compactness

round seats, not stringy ones

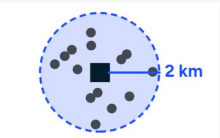
2 Polling logistics

Can every voter actually vote?



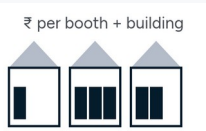
Station capacity

at most 1,200 electors per station



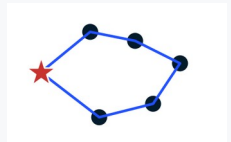
Walking distance

every voter within 2 km on foot



Station cost

fewest stations and premises



EVM transport

short drives from the dispatch centre

What We Have Done

4 districts modelled: Sangli, Satara, Nashik and Pune

52 Assembly seats redrawn, every one within ±10%

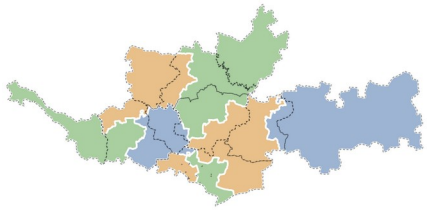
20,196 polling stations placed for 1.9 crore electors (2024 rolls)

4,773 buses and jeeps deliver 99.6% of EVM sets

Our Approach

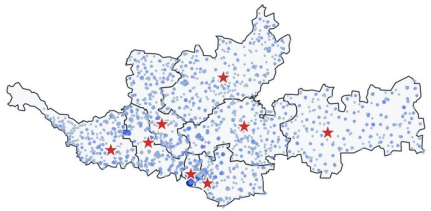
Every district runs through the same pipeline, from drawing seats to checking against today's elections

1 Draw seats



- **ReCom** search over connected maps
- 3 weightings × 2 seeds

2 Place stations



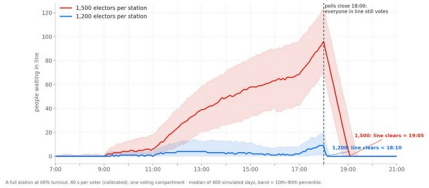
- **MIP** on the walk network
- ≤ 1,200 electors, ≤ 2 km walk

3 Route EVMs



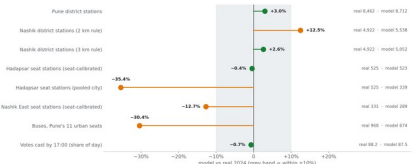
- **Multi-vehicle** routing from the dispatch centre
- Buses and jeeps on real roads

4 Simulate queues



- **Queue simulation** at each station
- Waiting time through polling day

5 Compare with today



- Against **today's seats** and **2024 figures**
- Same metrics, same network

Four Districts, Same Rules

Sangli

8	2.82 M	756
seats	people, 2011	units

Satara

8	3.00 M	1,767
seats	people, 2011	units

Nashik

15	6.11 M	2,475
seats	people, 2011	units

Pune

21	9.43 M	2,555
seats	people, 2011	units

Same rules everywhere: ±10% of the district's own average seat, connected seats, no thin necks; seats stay inside district lines. Population: Census 2011. Units: villages, towns and city wards.

Districting Model: ReCom

Seats are connected pieces of a border graph, improved by a guided random search

The Graph

- **Nodes**: villages, towns and city wards
- **Edges**: two units share a land border
- A **map** splits the graph into K connected seats

MUST PASS

- **Connected**: every seat is one piece
- **Population band**: within ±10% of target

$$0.9T \le P_s \le 1.1T$$

$$T = P/K$$ district population ÷ seats

- **No thin necks** between two large lobes
- Each final seat is **one polygon**

The Score

Weighted score, **lower is better**

$$J = w_{pop} L_{pop} + w_{dist} L_{dist} + w_{tal} L_{tal} + w_{shape} L_{shape}$$

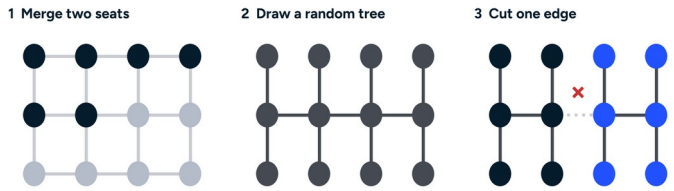
- **Population**: gap from the target seat size
- **Distance**: people to their seat centre
- **Taluka**: extra splits of a taluka
- **Shape**: compactness loss (Polsby-Popper)

$$L_{shape} = 1 - \frac{4\pi A_s}{\ell_s^2}$$

THREE WEIGHT PROFILES

Profile	Pop.	Dist.	Taluka	Shape
Balanced	0.35	0.30	0.20	0.15
Locality	0.30	0.45	0.10	0.15
Taluka	0.30	0.20	0.35	0.15

The Search



- **Merge** two neighbouring seats
- **Tree**: random spanning tree on real borders
- **Cut** one edge: two new connected seats

Simulated annealing: accept worse maps early, fewer later

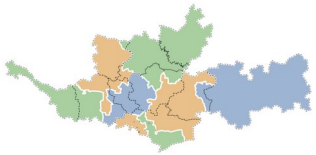
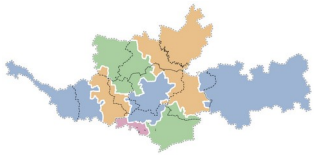
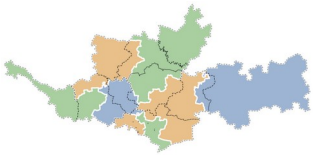
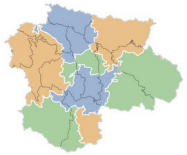
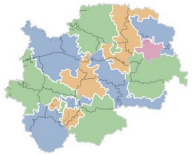
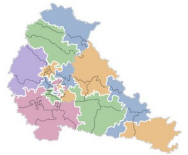
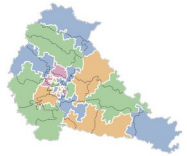
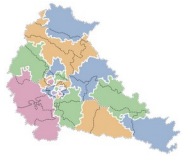
$$P(accept) = min(1, e^{-\Delta J/\tau})$$

- Temperature τ falls as the search runs
- **Best valid map** over all steps is kept

Research constraints, not a legal delimitation test. Every map is rescored on one common baseline before comparison.

Districting Results by Weighting

Three weightings per district, each the better of two seeds; the outlined map scores best on the common score

	Balanced	Locality	Taluka
Sangli	 max dev **4.0%** · taluka cuts **8** · score **0.59**	 max dev **1.9%** · taluka cuts **11** · score **0.65**	 max dev **2.1%** · taluka cuts **7** · score **0.56**
Satara	 max dev **1.1%** · taluka cuts **16** · score **0.80**	 max dev **1.6%** · taluka cuts **19** · score **0.88**	 max dev **3.5%** · taluka cuts **14** · score **0.77**
Nashik	 max dev **4.2%** · taluka cuts **26** · score **1.03**	 max dev **7.4%** · taluka cuts **27** · score **1.11**	 max dev **5.0%** · taluka cuts **22** · score **0.97**
Pune	 max dev **9.5%** · taluka cuts **30** · score **1.10**	 max dev **7.5%** · taluka cuts **38** · score **1.32**	 max dev **8.7%** · taluka cuts **29** · score **1.11**

Which Weighting Wins

District	Best map	Max dev	Taluka cuts
Sangli	Taluka	2.1%	7
Satara	Taluka	3.5%	14
Nashik	Taluka	5.0%	22
Pune	Balanced	9.5%	30

All 52 seats inside ±10% in every map; today 26 of 52 are outside.

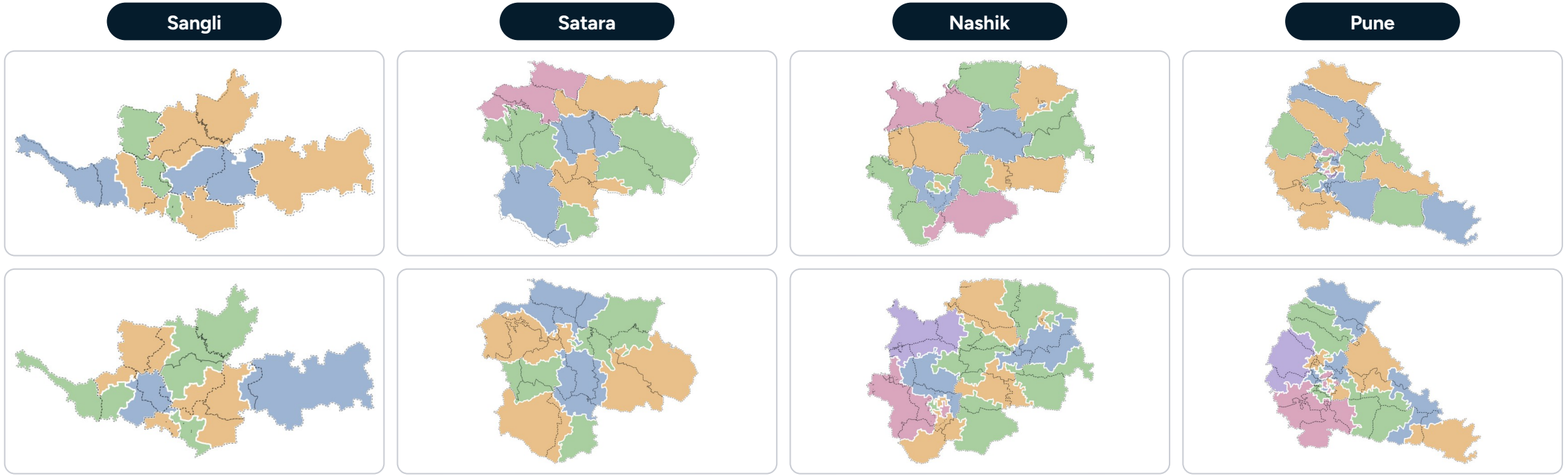
Taluka weighting scores best in three districts with the better seed, but the ranking changes with the seed (see the audit).

The price of balance: more taluka cuts than today, e.g. Pune 30 vs 18.

Lower is better for deviation, taluka cuts and score. Common score: one shared weighting of population, road distance to the seat centre, taluka cuts and shape. Census 2011 population; dashed lines = talukas.

Comparison with Current Constituencies

The 2008 delimitation and our best map, scored on the same Census 2011 units and metrics



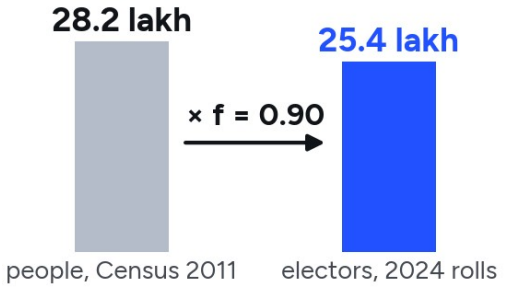
KPI (Census 2011)	Sangli		Satara		Nashik		Pune	
	Today	Ours	Today	Ours	Today	Ours	Today	Ours
Seats outside ±10%	2 of 8	**0 of 8**	2 of 8	**0 of 8**	7 of 15	**0 of 15**	15 of 21	**0 of 21**
Largest ÷ smallest seat	1.27×	**1.03×**	1.19×	**1.06×**	1.81×	**1.10×**	3.02×	**1.17×**
Max deviation from target	15.0%	**2.1%**	11.1%	**3.5%**	42.4%	**5.0%**	67.2%	**9.5%**
Extra taluka cuts	**5**	7	**13**	14	**11**	22	**18**	30
Avg. road km to seat centre	**15.9**	16.4	**17.9**	18.4	**15.4**	16.2	**11.3**	11.6

Highlighted: the better value in each pair; lower is better on every row. Boundaries: DataMeet 2008 polygons.

Polling Station Location: Data Pipeline

From census counts to candidate station sites in four steps

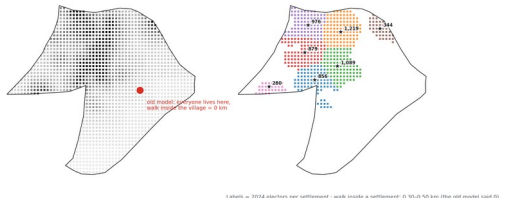
1 Count voters



Census population scaled to 2024 electors

- **2011 census** count per village
- × **seat factor** f = 2024 electors ÷ 2011 population
- District total matches the **2024 rolls**

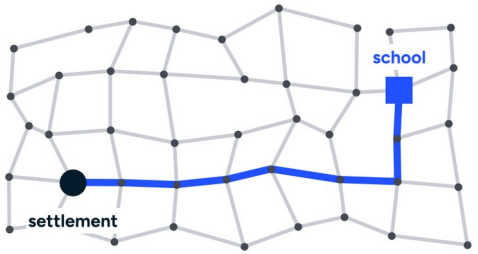
2 Find where people live



WorldPop 100 m cells grouped into settlements

- **WorldPop** 100 m × 100 m population grid
- Drop **farmland noise**, join nearby cells
- Each cluster becomes one **settlement point**

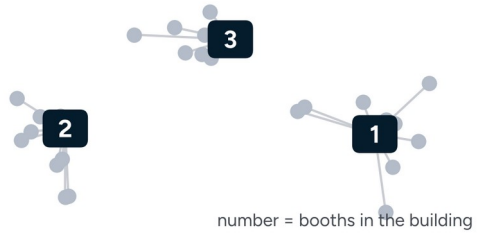
3 Measure walks



OpenStreetMap walk network

- **OpenStreetMap** roads, footpaths, tracks
- **Dijkstra** shortest walk to each candidate premises
- Keep only pairs within **2 km**

4 Optimise stations



Settlements assigned to station premises

- **Mixed-integer program** picks premises
- ≤ **1,200 electors** per station
- One **dispatch centre per seat**, drive cost included

100 m population grid

2 km maximum walk

1,200 electors per station

Step 1 shows Sangli: electors are fewer than people (children, migration), so f is below 1. Data: Census 2011, ECI 2024 elector rolls, WorldPop 100 m population, OpenStreetMap. Step 2 shows Ving village (Karad South): 100 m cells (darker = more people) become 7 settlements.

Polling Station Location: MIP Formulation

A mixed-integer program chooses premises, stations per premises and one EVM dispatch centre per seat

Decision Variables

$y_j \in \{0,1,\ldots,12\}$	stations opened at site j
$o_j \in \{0,1\}$	premises j is used
$x_{ij} \geq 0$	electors from settlement i sent to site j
$z_k \in \{0,1\}$	dispatch centre k is chosen
$f_{kj} \geq 0$	stations at j supplied from centre k

DATA

v_i	electors in settlement i
w_{ij}	walking km, settlement i to site j
d_{kj}	driving km, centre k to site j

Objective: Minimise

$c_B \sum_j y_j$	stations opened
$+c_O \sum_j o_j$	premises used
$+c_W \sum_{i,j} w_{ij} x_{ij}$	total walking
$+c_E \sum_{k,j} d_{kj} f_{kj}$	EVM transport

COST WEIGHTS

c_B	c_O	c_W	c_E
25,000	**5,000**	**10**	**120**

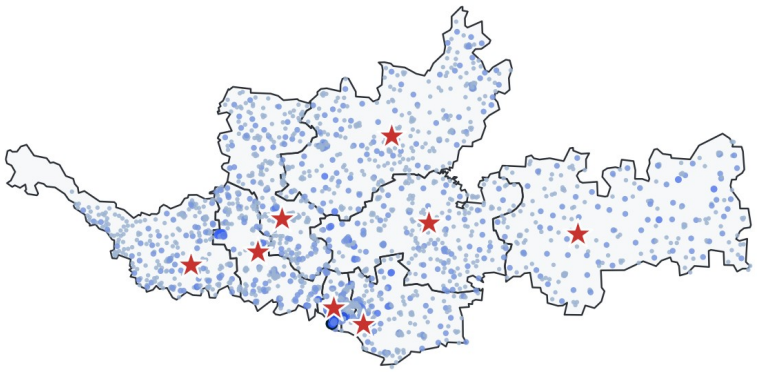
Constraints

$\sum_j x_{ij} = v_i$	every elector assigned
$x_{ij} = 0 \; if \; w_{ij} > 2\,km$	walks ≤ 2 km (own settlement always allowed)
$\sum_i x_{ij} \leq 1200\, y_j$	station capacity
$y_j \leq 12\, o_j$	stations only in open premises
$\sum_k z_k = 1$	exactly one dispatch centre
$\sum_k f_{kj} = y_j$	stations at j all supplied
$f_{kj} \leq 12\, z_k$	only the chosen centre supplies

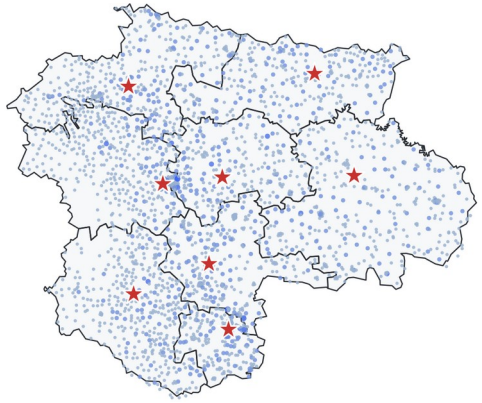
Polling Station Location: Results

Optimised plan on 2024 rolls and today's seats, with at most 1,200 electors per station and a 2 km walk

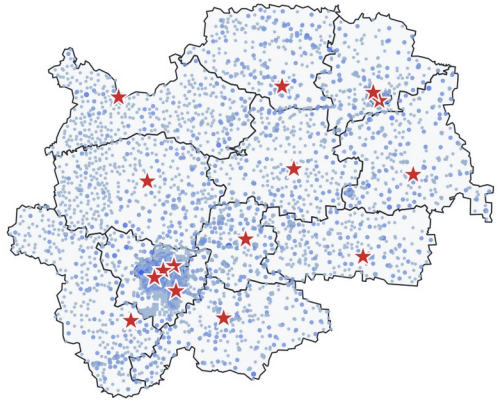
Sangli · 2,669 stations



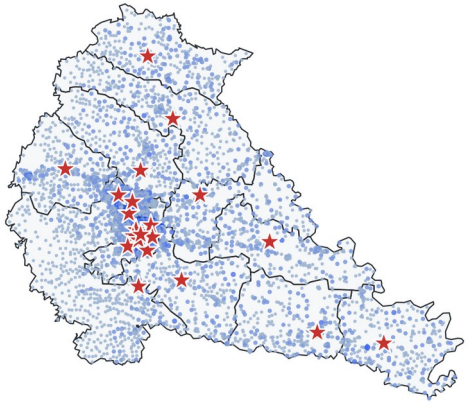
Satara · 3,277 stations



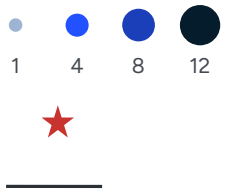
Nashik · 5,538 stations



Pune · 8,712 stations



Legend



Polling premises: size and shade = stations inside (1 to 12)

EVM dispatch centre (one per seat)

Seat boundary (today's 2008 seats)

District	Stations	Premises	Electors / stn	Avg walk	Over 2 km
Sangli	2,669	1,964	953	0.54 km	0.46%
Satara	3,277	2,652	810	0.62 km	0.43%
Nashik	5,538	4,149	915	0.59 km	0.35%
Pune	8,712	5,767	1,016	0.50 km	0.36%
All four	20,196	14,532	946	0.54 km	0.38%

About half a kilometre is the average walk in every district, counted from each settlement, not the village centre.

Fewer than 0.5% of electors walk over 2 km, and only because their own settlement is spread out: its centre alone is over 2 km away.

Gurobi facility-location model per seat, on 2024 rolls; premises at settlement centres from the WorldPop 100 m grid; walks on OpenStreetMap paths.

Election Logistics: Movement and Cost Heads

How EVMs and polling teams move before and after the poll, and the cost heads a district has to plan for

1 How EVMs and polling teams move

PRE-POLL

EVMs are stored, checked and allocated to constituencies, then randomised to polling stations. On dispatch they travel with polling teams from the election office or distribution hub to the stations. Districts arrange vehicles and may track routes.

POST-POLL

Sealed EVMs travel from polling stations to collection centres and secure storage.

2 Cost areas to quantify

Transport	vehicle hire, fuel, driver costs, distance, dispatch and collection trips, fleet size
People	polling-team and support-staff numbers, duty payments, training, staff transport
Sites and equipment	station setup, premises, EVM/VVPAT preparation, batteries, paper rolls, storage
Security and field support	security deployment, sector officers, reserve-equipment response

What a Real Report Shows: Madurai South, 2016 RTI Report

₹41.03 lakh
total reported spend

₹25.87 lakh
staff remuneration

₹6.69 lakh
vehicles: fuel and hire

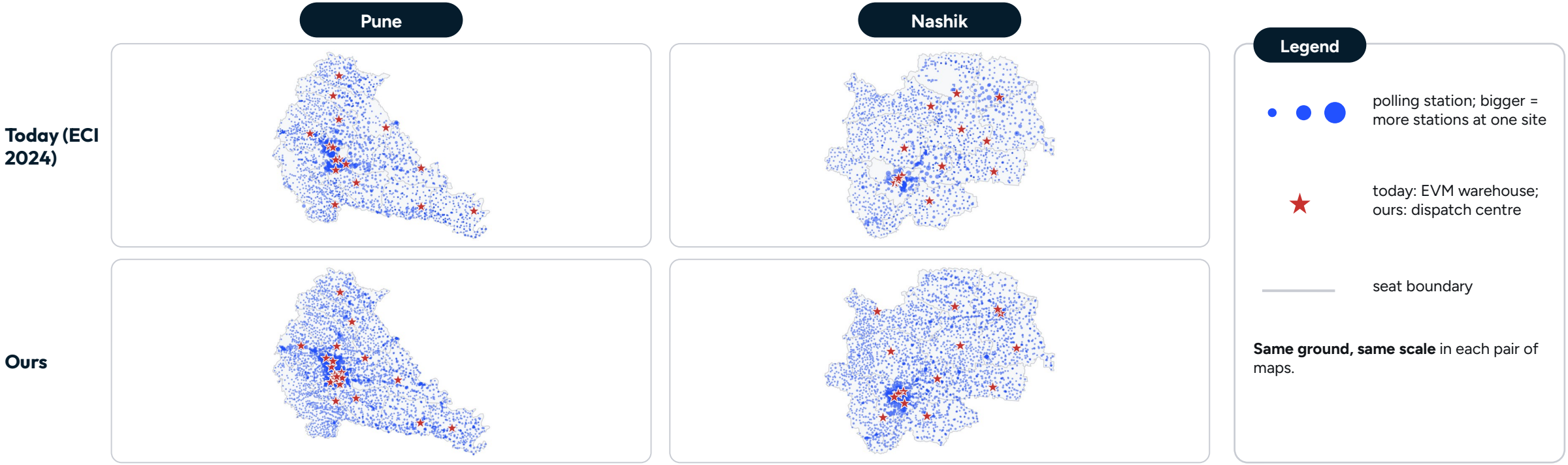
56 · 2,418
polling stations · workers

The report questioned some records as incomplete. An example of cost categories, not a benchmark for Maharashtra.

Sources: ECI EVM guidance; Pune transport arrangements (2024); Times of India, Madurai South.

Our Plan vs Today's Stations and EVM Warehouses

Real 2024 ECI polling stations and the warehouses that supply them, against the model's stations and dispatch centres



	Pune today	Pune ours	Nashik today	Nashik ours
Polling stations	6,582	8,712	4,191	5,538
EVM warehouses / dispatch centres	16	21	15	15
Total km for today's station count (one trip per station)	119,391	**91,622**	86,244	**72,105**
Average km per station	18.1	**13.9**	20.6	**17.2**
Stations within 20 km of their warehouse	70%	73%	60%	62%
Vehicle-km with shared bus and jeep routes, out and back	not known	116,577	not known	96,454

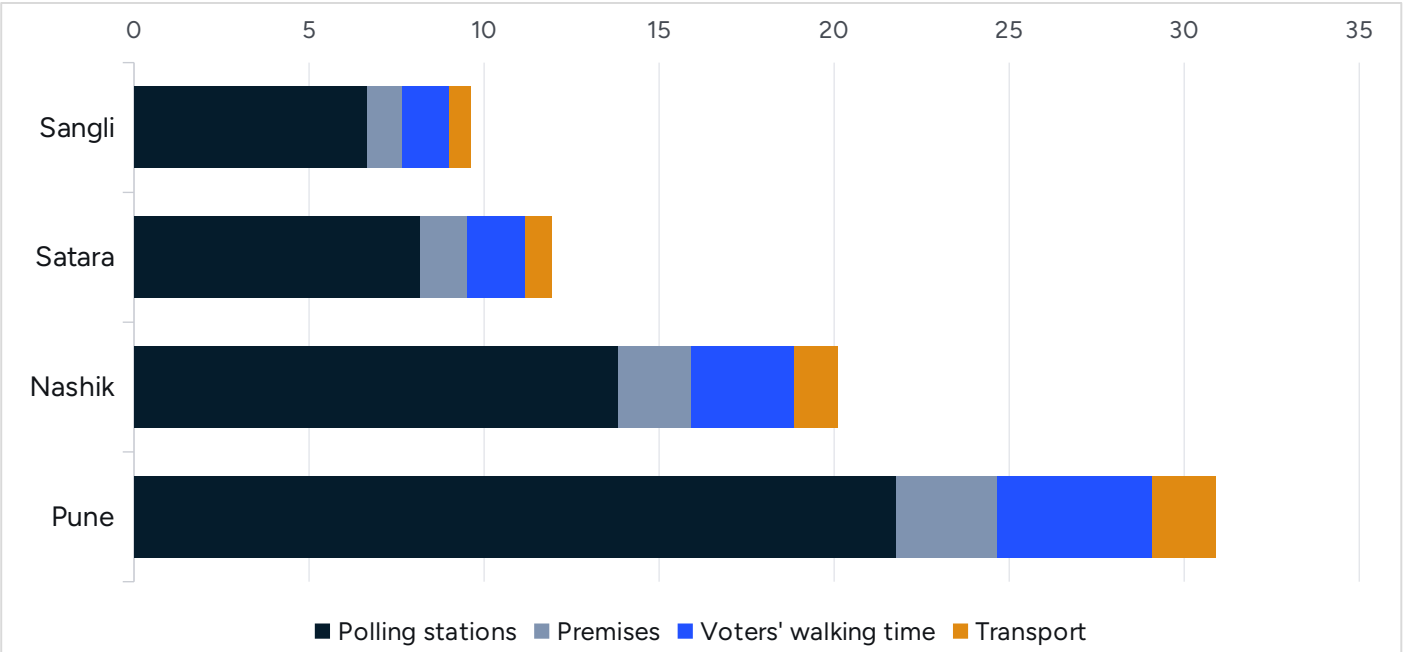
Logistics Cost of Our Plan

The two stages the model optimises, and what the resulting plan costs in each district

Strong room	Dispatch centre	Polling station	Receipt centre	Strong room
EVMs stored and randomised, district HQ	teams collect EVM sets, one centre per seat	sets reach stations the evening before	sealed EVMs returned on polling night	EVMs held until counting

Dark stages are optimised by the model: where the dispatch centre sits, and how buses and jeeps reach every station.

Cost by district (₹ crore)



District	Total	Per elector
Sangli	₹9.6 cr	₹38
Satara	₹11.9 cr	₹45
Nashik	₹20.1 cr	₹40
Pune	₹30.9 cr	₹35

Cost head	Unit cost used
Polling station	₹25,000 each
Premises	₹5,000 each
Voters' walking time	₹10 per elector-km
Bus (6 EVM sets)	₹9,000 hire + ₹35 per km
Jeep (1 EVM set)	₹3,500 hire + ₹14 per km

Plan on 2024 rolls and today's seats, 1,200 electors per station. Station cost covers the polling team of four, training, furniture and EVM share. Strong-room and counting costs are outside the model.

EVM Dispatch Routing: Model

Which vehicles should go where, in what order, so every set arrives on time at the lowest cost?

1 What has to move



polling team of 4 — one EVM set

travel together to one station

One EVM set and its team per station

- **EVM set:** control unit, ballot unit, VVPAT
- **Polling team** of four travels with it
- Collected at the **dispatch centre** of the seat

2 Vehicles



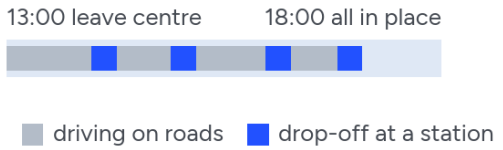
bus: 6 EVM sets — **jeep: 1 set**

the solver chooses the mix

Buses for bulk, jeeps for outlying hamlets

- **Bus:** 6 sets, ₹9,000 hire + ₹35 per km
- **Jeep:** 1 set, ₹3,500 hire + ₹14 per km
- A premises with more than 6 stations **gets several stops**

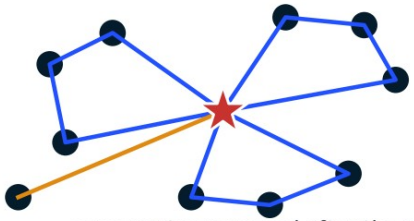
3 Time window



Drive times on OpenStreetMap roads

- Leave at 13:00, **in place by 18:00**
- **Each stop:** 10 min + 3 min per set unloaded
- Times from **real road speeds**

4 Solve the routes



same routes reversed after the poll

Vehicle routing with time windows (OR-Tools)

- Minimise **hire + kilometre cost**
- **One problem per seat**, from its dispatch centre
- **Same routes reversed** to bring sets back after the poll

20,196 EVM sets to deliver across the four districts, one per station

5 h from 13:00 departure to 18:00 arrival for every set

52 routing problems, one per seat and its dispatch centre

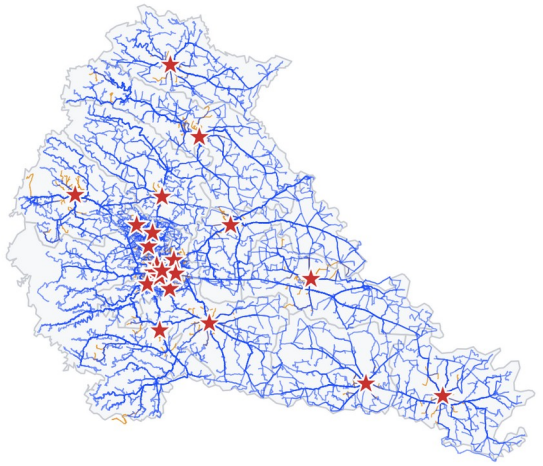
EVM Dispatch Routing: Results

Every bus and jeep route on the eve of the poll, from each seat's dispatch centre

Sangli · 8 seats



Pune · 21 seats



bus route (≤ 6 EVM sets) jeep route (1 set) dispatch centre

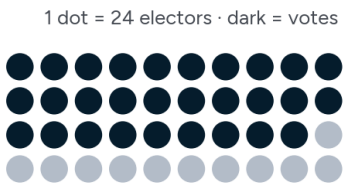
	Sangli	Pune
EVM sets to deliver	2,669	8,712
Buses + jeeps	400 + 272	1,346 + 685
Vehicle-km, out and back	45,587	116,577
Last set in place	16:48	17:54
Sets not routed	0	41

Shared routes cut vehicle-km by 77% against one trip per station from the largest town, and choosing the dispatch centre saves another 14%: from 1,630k to 324k km across all four districts.

Polling Day Simulation: Model

How we simulate polling day at every station the plan opens, voter by voter

1 Voters at each station

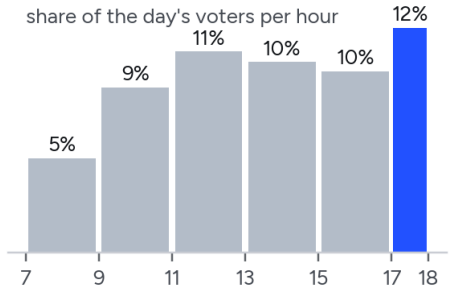


Sangli average: 953 electors × 73% = 692 voters

Electors × the seat's 2024 turnout

- **Electors** from the station plan, at most 1,200
- **Turnout** of the station's seat in the 2024 election
- Drawn at random **each simulated day**

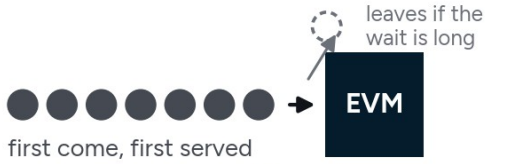
2 When they arrive



Maharashtra's reported 2024 turnout by hour

- **Arrival times** follow the 2024 turnout curve
- Slow start, **steady from 11:00**, last-hour rush
- Each voter gets **a random time** on that curve

3 Queue at the compartment

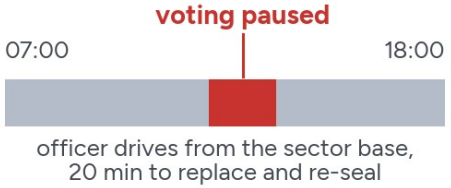


40 s each, on average

One voting compartment, first come first served

- **Voting time** varies, 40 s on average
- Voters may leave **if the wait passes 20 min**
- Everyone in line at 18:00 **still votes**

4 EVM failures



A failure pauses voting until a spare arrives

- **10% of stations** lose a unit (Assam 2024 rates)
- **Sector officer** drives a spare over; 20 min to fit
- No spare left: **fetched from the dispatch centre**

30 simulated polling days for every station, so results are averages, not one lucky day

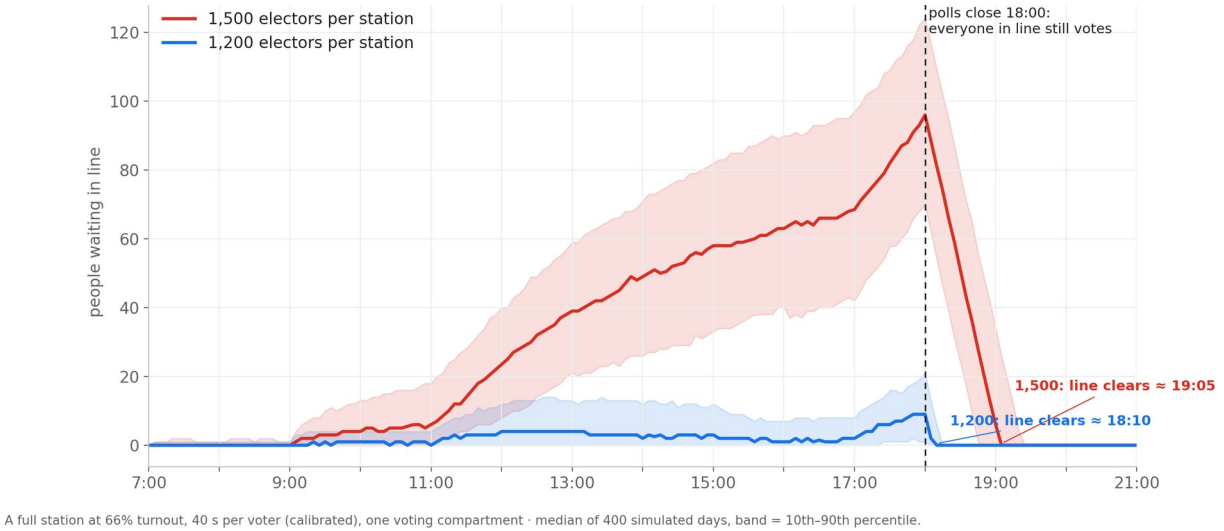
40 s average voting time: the value whose turnout curve best matches 2024

1,200 vs 1,500 electors per station: the two caps we compare

Polling Day Simulation: Results

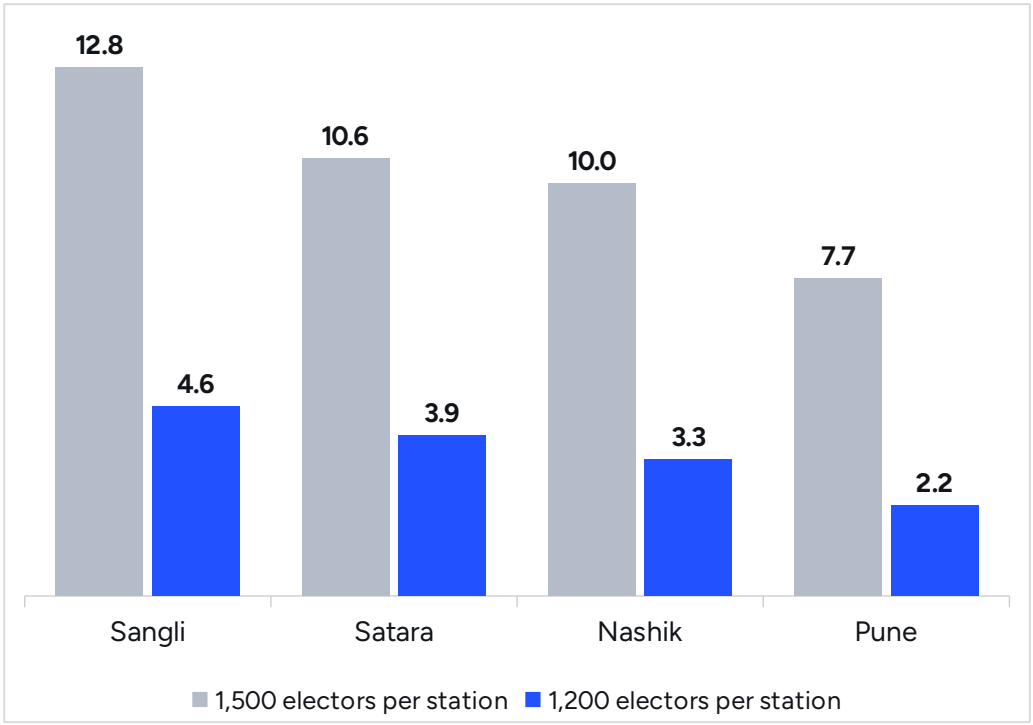
ECI cut the cap from 1,500 to 1,200 electors per station in 2025: what that does to queues on polling day

One full station through the day



A full station at 66% turnout, 40 s per voter (calibrated), one voting compartment · median of 400 simulated days, band = 10th–90th percentile.

Average wait per voter (min)



5 to 16% of voters wait over 30 min at 1,500 per station; under 1% at 1,200

₹4 extra per elector for the stations the 1,200 cap needs

±1 point gap between simulated and reported 2024 turnout at every hour

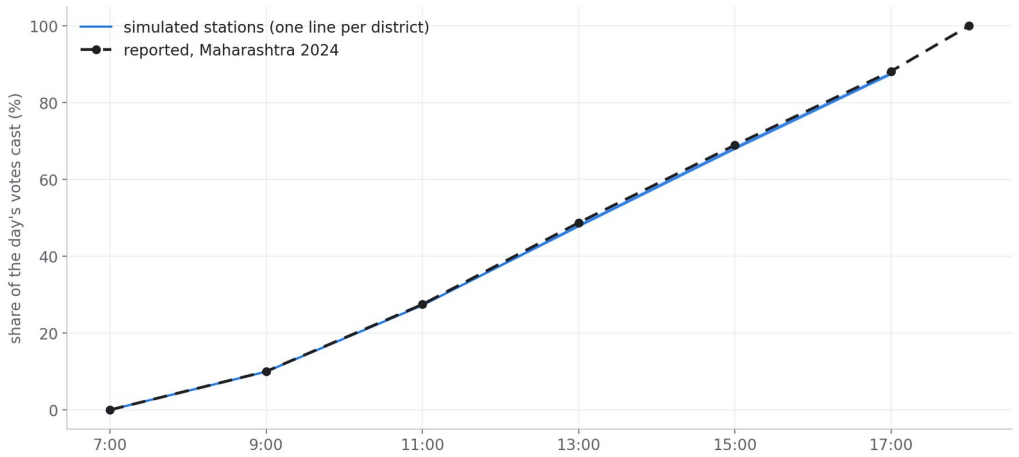
Validation Against the 2024 Election

Model outputs compared with every public 2024 figure we found for these districts

Check	Real 2024	Model	Error
Polling stations, Pune district	8,462	8,712	+3.0%
Electors per station, Pune	1,046	1,016	−2.9%
Polling stations, Nashik district	4,922	5,538	+12.5%
Electors per station, Nashik	1,030	915	−11.2%
Stations, Hadapsar seat	525	523	−0.4%
Stations, Nashik East seat	331	289	−12.7%
City buses, 11 Pune seats	968	674	−30.4%
Votes cast by 17:00	88.1%	87.7%	−0.4 pts

Where it misses: Nashik's rural seats open more stations than reality (+12.5%), and Pune's city fleet is one of several the district used, so the model needs fewer buses.

Polling day: simulated vs reported turnout



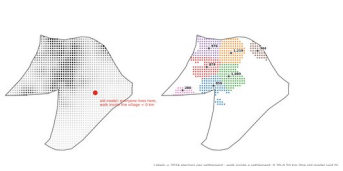
Calibrated at 40 s per voter (best fit: 0.6 points mean error; 50 s: 1.3; 60 s: 2.0). Reported: Maharashtra CEO via Deccan Herald, The Wire.

Green: within 5%; amber: within 15%; red: further off. Sources: Deccan Herald, The Week (PTI), Punekar News, Maharashtra CEO. Sangli and Satara district station totals were not found in public sources.

Summary

Each stage of the pipeline and its main result across the four districts

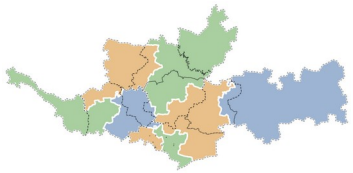
1 Data



7,553 villages

and city wards, from Census 2011, 2024 rolls, WorldPop and OpenStreetMap

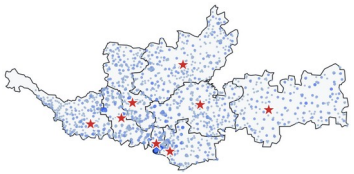
2 Seats



52 seats

all within ±10% of target; today 26 are outside

3 Stations



20,196 stations

99.6% of electors within 2 km on foot

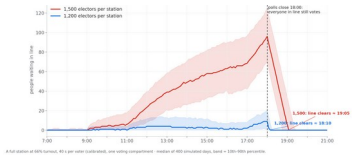
4 Transport



4,773 vehicles

buses and jeeps deliver 99.6% of EVM sets by 18:00

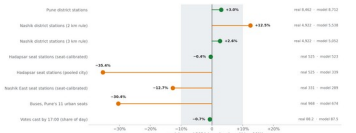
5 Polling day



3 min wait

average per voter at 1,200 per station (2 to 5 min by district); under 1% wait over 30 min

6 Validation



3% gap

between modelled and real 2024 polling stations in Pune

What It Adds Up To

Fair seats by construction

Every map ReCom visits is connected and inside the population band; the search only trades off distance, talukas and shape.

Stations people can walk to

Real electors, real settlements and real footpaths replace one point per village and straight lines.

One pipeline, any district

Seats stay inside district lines, so the same models scale from four districts to all 36 in Maharashtra.